

Chapter 3: Second Review

3.1 UML design guidelines

To make life easy it is advisable to start with a class diagram before you do any other diagrams as the class diagram will guide you in coming up with all other diagrams.

3.2 Class diagram

The class diagram shows the static view of the system

3.2.1 How to identify classes:

One can use the following to come up with potential classes:

The classes are derived from your proposed solution

1. Tangible objects

These are the objects that we see like vehicles, buildings, cars and others

2. Roles

People play different functions in an organization and this can be a source for our classes, like employees, members of the public or even guests

3. Events

The events that can happen in an organization like a room being booked, car being hired, ticket being reserved, book being borrowed or being returned, an order being made and others

4. Organizational units

There are the departments that we can have in an organization like Accounts, Marketing, Technical, Manufacturing and others

3.2.2 What is a class diagram?

This is basically a group of boxes showing classes and the associations that do exist between each class and the other. The class represents all the information that we have in the problem domain.

Example:

Let us consider a library system where members can include students and staff members who can borrow books up to a maximum of 5 at each borrowing. There is a receptionist who adds members to the system and captures the books to be borrowed by each member. A library administrator maintains the library system.

3.2.3 Multiplicities

Here we are saying there is a limit to a number of books that a member can borrow. A member can borrow 1 book up to a maximum of 5 books. So the multiplicity is 1..5. but however if a member is allowed to borrow just one book the multiplicity will be 1. Let us look at another example, if the member is allowed to take as many books as they like then we use this *.

3.2.4 How to select candidate classes for a Class diagram?

From the proposed solution list all nouns that you feel can qualify to be called classes: Let us consider the following examples: Now let us use the following concepts we mentioned at the top:

Tangible objects; Roles; Organizational units and Events to come up with potential classes

Potential classes	Keeping or Discarding
Library	Will keep. We are creating a Library System
Librarian	Will discard: No information will be kept about the Librarian for now
Member	Will keep. To keep the member's information
Book	Will keep. To keep the book details
Library Administrator	Will discard: No information will be kept about the Administrator for now

So we have kept these following classes: Library, Member and Book

3.2.5 Associations

The association is the link that exists between one class and the other like the connections that may exist between two classes over a period of time

A member is related to a Library because a member can use the library.

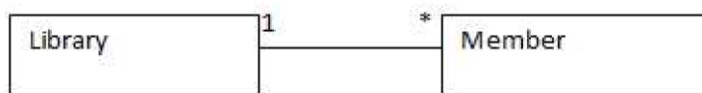
A member is related to a Book because a member can borrow a book from the library.

A book is related to the library because the book belongs in the library.

We can now come up with associations:

Association	Definition
Member Library	A Member can use the Library
Book Library	A book belongs in a library
Member- Book	A member can borrow a book in a library

Let us look at our: Library example: Class diagram



Multiplicity explanation: A library can have many (*) members but a member can only belong to one (1) library.

Your task is to draw the class diagram showing the association between the Library and the Book and together with the Member. Show correct multiplicities.

3.3 Object diagrams

The Object diagram is derived from the Class diagram and the diagram shows the state of the class diagram at a particular instance of time like when a member borrows 3 books from the library. This will be shown as objects which will be underlined. The object diagram should show the library objects like library1: Library, member name like peter: Member and the books being borrowed book1:Book, book2:Book, book3: Book. The book objects will all be connected to the library1 object and all the book objects will also be connected to the peter object since the member is borrowing the books. Peter object will be linked to the library1 object.

Your task is to represent the above using an object diagram

3.4 Sequence Diagram

A sequence diagram shows the interactions that happen between objects. A sequence diagram should capture a single scenario in a sequence diagram for example you can show a member borrowing a book from the library. If you wish to show another scenario you can come up with another sequence

diagram like when the member returns the book in the library. If you show the two scenarios in the same sequence diagram the diagram will become too complicated. To draw the sequence diagram you take the classes defined in the class diagram that help in showing the scenario that you want to represent. Each class should start with the object name and then separated with a colon then the class name, for example library1: Library and should be underlined.

Let us consider our library example:

Here we expect the user interface or an actor who could be the Librarian to initiate the borrowing process. The first message will be send to the library1 object which falls in the Library class and the message will be borrowBook(memberID). The library1 object will send the message identifyMember(memberID) to itself to check if the member is registered in the library to borrow books. This will result in an overlaid bar on our activation bar because the message is being send to the same object. The library1 object will send another message to itself again to find the book being borrowed and the message is findBook(bookID) to see if the book is available and we have an overlaid bar on our activation bar. If the book is available we can include a guard saying [book available] and then we remove the bookID from the book1 object which is of class Book; to show the book is no longer available in stock. We removeBook(bookID) from the book1 object, so the message will be send to the book1 object. The message removeBook(bookID) is being sent to the book1 object and this is an overlaid bar because the message is being sent to the same object to remove the book1 from the book class. We setloan(bookID) to the member1 object.

We have something like this:



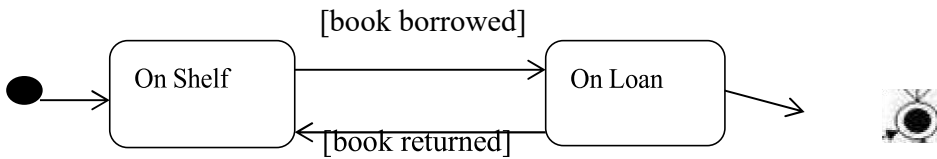
Complete the sequence diagram using the above information.

3.5 Communication diagram

It is easy to do a sequence diagram first before you do the communication diagram. The communication diagram is just like a sequence diagram but here we are just numbering the messages as they communicate with other objects. For example we have our userInterface sending 1: borrowBook(memberID) to library1 object. The library1 object will send the message 1.1: identifyMember(memberID) to itself and then send 1.2: findBook(bookID) to itself again. The process goes on like that. The crucial thing to remember is that every message in a communication diagram should be numbered.

3.6 State charts

The state chart basically shows how an object changes its state in response to events that are happening around the object. It is crucial to identify the states and events when coming up with a state chart diagram. A book in a library can have two states on the shelf or borrowed. The action that will cause the state to change from one state to the other will be put in guards like [book borrowed] and [book returned]. We also have the initial state and the final states in the state chart.



A more complicated state chart could show something like this:

A book can be On Shelf and then a member places the book on reservation so the book will be in another state like On Hold. If the member later cancels the reservation and the book gets back to the On Shelf state. If the member decides to borrow the book, the book will move its state to On Loan. The book will later be returned and gets back to the On Shelf state.

3.7 Process Design

Making use of <<precondition>> and <<postcondition>> to handle a successful operation or using the State chart diagram or a Component diagram, Task analysis

- Pre-condition: what you assume to be true in order for the use case to succeed Post-
- condition: what the use case will have be accomplished

You need to pick one operation that you need to show using a pre and post condition:

Let us consider an operation for a member borrowing a book from the library:

Context LibraryExample:: borrowbook (library1:Library, book:Book, memberID:Member, loanDate:Date, returnDate: Date)

Pre

- loanDate should be before returnDate
- book is one of the books in the library
- memberID is registered in the library

Post

- a new Loan object will have been created

--LibraryExample will be linked to memberID

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Complete the other post conditions

3.8 Networking Projects:

For networking we expect the following

For a network project you either come up with a situation where there is need to come up with a design and an implementation of an Enterprise network from scratch, the other option is to propose improvements and or expansion of existing enterprise network. The document should contain but not limited to the following:

- Logical and Physical diagrams

As part of your design make sure you both have logical and physical designs which can be achieved with the aid of packet tracer, vision or any tool which you are comfortable with.

- The Hierarchical Network Model

Since you will be designing a network worth a project you are mostly likely to use a hierarchical Network Model which divides you enterprise network into 3 layers; core, distribution, and access layer.

- VLANs design

This is a sub part of your logical design where you are going to suggest a network address which you will subnet to form subnets. You are mostly likely to use Virtual LANs instead of the physical LANs and you should be able to justify every design decision you make.

- Subnetting using Variable Length Subnet Mask

As part of your subnetting, since the organization which you will be designing the network for is likely to have departments which vary in sizes in terms of employees working in those departments, you will be required to implement subnetting using VLSM.

